

As you are aware, broadly speaking, GNP is made up of consumption, saving exports net of imports, besides net factor income from abroad. Saving is important in as much as it goes to finance investment, which in turn brings about growth of GNP. What is the volume of Saving relative of GNP? How much of it is consumed by the needs of depreciation? Who all contribute, and how much, to the total volume of Saving? Let us see what kind of data is available on such questions.

Estimates of Gross Domestic Saving (GDS) and Net Domestic Saving (NDS) in *current prices* and the Rate of Saving are made by CSO and published in the **National Accounts Statistics (NAS) every year** and the **Press Note of January of every year releasing First Revised Estimates**. These are first made for any year along with First Revised Estimates of GDP, etc., and revised and finalized along with its subsequent revisions. The structure of Savings, that is, the distribution of GDS and NDS by type of institution – household sector, private corporate sector and public sector are also available in this publication. Also presented in this document is the distribution of:

- i) GDS, NDS and consumption of fixed capital (depreciation) by public sector, private corporate sector and household sector;
- ii) Public sector GDS, consumption of public sector fixed capital and public sector NDS by type of institutions²²;
- iii) Public sector GDS – by public authorities [sub-divided further into government administration and departmental (commercial) enterprises], non-departmental enterprises (broken up further into government companies and statutory corporations, the latter including port trusts also);
- iv) Private sector GDS, consumption of fixed capital and NDS by private corporate sector and household sector;
- v) Private corporate sector GDS by joint stock companies (distributed further into financial and non-financial companies) and cooperative banks and societies;
- vi) GDS of the household sector into (net) financial saving and saving in physical assets and that of (net) financial saving, that is, currency, net deposits, shares and debentures, net claim on Government, life insurance funds and provident and pension funds;
- vii) Financial assets [in greater detail than in (vi) above] and liabilities of the household sector leading to net financial saving of the household sector; and
- viii) Item (xiii) in the list in sub-section (a) of Section 20.3.2 – these indicate estimates of net saving in these entities.

NAS 2014 presents the above estimates for 2012-13 and for the period 2004-05 to 2012-13. The **Press Note dated the 31st January, 2014 of MoSPI** releasing **Quick Estimates** provides estimates of GDS, NDS, and all the estimates referred to at (i) to (vii) above for 2012-13. Time series of estimates of GDS and NDS in *current prices* are, however, available from 1950-51 onwards. The

²² See footnote 12.

time series on the structure of GDS by institutions (public, private corporate and household sectors) are presented in the **Pre-Budget Economic Survey (2014)** of the Ministry of Finance. **The Handbook of Statistics on the Indian Economy of RBI** presents time series data from 1952-53 on GDS and NDS and from 1970-71 onwards on (i) their components by institutions, and (ii) financial assets and liabilities of the household sector, item-wise, at current prices. **The EPWRF publication on NAS (fifth edition)** gives time series data from 1950-51 on all these variables and on Domestic Saving in public sector by type of institutions. The CMIE publication referred to earlier also contains time series data on GDS and NDS, though for a shorter period of time.

Estimates of Gross and Net Domestic Saving at the State and Union Territory levels are not being made at present by DESs of State/Union Territories.

The CSO publishes “**Sources and Methods**”, an additional publication showing the detailed methodology followed and data sources used for preparation of the National Income aggregates in the base year and the methodology to be followed since the change in base year. It may be added that estimates of Savings suffer from a number of limitations due largely to deficiency of data. This is particularly so in respect of estimates of various financial instruments from the private corporate sector and the household sector. Saving being the excess of income over expenditure, the major gaps in data relate to the household sector, non-profit institutions serving households (NPISH) and local bodies. Quality of data relating to the private corporate sector poses an additional problem. The estimate of Savings of the household sector in physical assets is thus derived as a residual, in the absence of direct data.²³

21.5 INVESTMENT

Investment is Capital Formation (CF). Investment of money in the shares of a company or purchase of land is not investment but buying a house or machinery is investment. In other words, investment is creation of physical assets like machinery, equipment, building, inventories and so on, called “produced assets”, adding to the capital stock (of such assets) in the economy, enhances the productive capacity of the economy. Investment or CF is another important component of GNP and the rate of investment – expressed as a proportion of GNP – largely determines the rate of growth of the economy. How is capital formation financed by the economy? What is the contribution of different sectors to capital formation or, how much is used up by different sectors? What is the capital stock available in the economy? These are all the questions that rise in one’s mind when considering strategies for economic growth. What kind of data is available?

The annual NAS publication of the CSO presents estimates of Gross Domestic Capital Formation (GDCF), Gross Domestic Fixed Capital Formation (GDFCF), Change in Stocks, Consumption of Fixed Capital (CFC), Net Domestic Capital Formation (NDCF) and Net Domestic Fixed Capital Formation (NDFCF) in current prices and at constant prices. These estimates are made along with First Revised Estimates of National Income in January every year and the process of revision of these estimates proceeds along with that of the estimates of national income aggregates. The estimates of capital formation are presented in the NAS through two alternate approaches, namely,

through the flow of funds approach and through the commodity flow approach. The flow of funds approach is gross domestic savings plus net capital inflows from abroad, while the commodity flow approach, derived by the types of assets. **The successive issues of NAS and the Press Note of MoSPI of January every year** also present estimates of the distribution of the following aggregates at current and constant prices:

- i) GDCF, GDFCF, Change in Stocks, CFC, NDCF and NDFCF by type of institutions (public sector, private corporate sector and household sector);
- ii) GDFCF in the public sector, private corporate sector and household sector by type of assets (construction and machinery & equipment);
- iii) GDCF, GDFCF, Change in Stocks, CFC, NDFCF and NDCF by economic activity²⁴;
- iv) Net Capital Stock (NCS), Net Fixed Capital Stock (NFCS) and Inventory as on the 31st march of each calendar year by type of institutions (Government administrative departments, Departmental enterprises, Non-departmental enterprises, joint stock companies of the private corporate sector, cooperative and household sector);
- v) NCS, NFCS and Inventory as on the 31st March of each calendar year by economic activity;
- vi) Consolidated Account of the Nation – How Capital Formation is financed (current prices);
- vii) Consolidated Account of the Nation – External Transactions showing current and capital transactions (in current prices);
- viii) GDCF, GDFCF and Change in Stock in the public sector by economic activity and type of institutions;
- ix) NDCF, NDFCF and Change in Stock in the public sector by economic activity and type of institutions;
- x) GDFCF in each type of public sector institutions by type of assets (at current prices);
- xi) (v) above for the public sector;
- xii) Financing of Capital Formation in (a) Railways, (b) Public Sector Communication and (c) Administrative Departments including Departmental Enterprises other than (a) and (b), (d) Non-Departmental Financial and Non-Financial Enterprises (in current prices); and
- xiii) Time series of GDCF and GFCF from 1951 to latest financial year.

NAS 2014 presents estimates of aggregates listed above at current and constant (2004-05) prices for the period 2004-05 to 2012-13. The **MoSPI Press Note of the 31st January, 2015** provide estimates of GCF, GFCF and consumption of fixed capital in current and constant (2011-12) prices as also of the distribution of these by industry of use and by institutional sectors of the economy, as defined in the SNA, namely, Public non-financial corporations, private non-financial corporations, public financial corporations, private financial corporations, general Government and Households for the period 2011-12 to 2013-14.

²⁴ See footnote 10.

Sl. No.	Publication		Variable	Time Series From
1)	Pre-Budget Economic Survey (2006)		#GDCF, GFCF and Change in Stocks and their structure by by institutions	1950-51
2)	RBI Handbook of Statistics on the Indian Economy – 2005	*i)	GDCF and NDCF	1950-51
		*ii)	GDCF, GFCF and Change in Stocks and their structure by institutions.	1970-71
		#iii)	Financial assets and liabilities of the household sector (item-wise)	1970-71
3)	EPWRF publication on NAS (Fifth Edition)	*i)	data of the type in items (i) to (ix) and (xi) of NAS listed in the preceding paragraph of this section are generally given.	1950-51
		*ii)	Capital-output ratios by institutions;	1950-51
		*iii)	Average NFCS to Output	1950-51
4)	CMIE		*GDCF and GFCF	1960-61

at current prices

*at current and constant (1993-94) prices

The publications/documents of **CSO** and other documents like the **Report of the National Statistical Commission** referred to in the section on National Income contain the methodology and the data utilized for preparing estimates of Capital Formation and related aggregates also.

Estimates of GFCF at the State level are being prepared by 14 States, namely, Punjab, Haryana, Himachal Pradesh, Madhya Pradesh, Orissa, Meghalaya, Assam, Andhra Pradesh, Tamil Nadu, Kerala, Karnataka, Maharashtra, Gujarat and Rajasthan. The status of these estimates are indicated in the **EPWRF publication on NAS**, referred to in the sub-section 20.3.4.

Gaps in data for the estimation of capital formation exist in a number of relevant areas²⁵. Some of these are:

- Non-availability of regular data on equipment and machinery (part of GFCF) in domestic production in unregistered manufacturing;
- Lack of annual enterprise surveys and surveys on NPISHs for estimating GFCF by institutional sectors of the economy;
- Complete accounts of local bodies are not available for some States/ UTs;
- Bank advances and other indirect approaches form the basis of estimation of Changes in Stocks, with benchmark estimates coming from quinquennial enterprise surveys; and

- v) Inadequacies in (a) the representative character of the sample of companies and (b) the blow-up factor used for estimation of saving and investment of the private corporate sector.

Check Your Progress 3

- 1) What approaches are followed by CSO to compile the saving data in National Accounts Statistics? Which estimates of investment are prepared by the CSO?

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- 2) Go through the methodology contained in NAS 2015 and the data utilized for preparing estimates of capital formation and related aggregates.

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- 3) From where can you get the data on contribution of different sectors to capital formation?

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21.6 LET US SUM UP

The Indian Statistical System with the **Ministry of Statistics and Programme Implementation (MoSPI)** at its apex collects, compiles and disseminates an enormous amount of data on diverse aspects of the Indian economy, to the extent feasible, international standards prescribed by the **United Nations** and other international bodies. **MoSPI** sets standards to be followed by statistical agencies in the country and coordinates statistical activities carried out by these agencies. It also develops standardized classification of economic activities, namely, **NIC**, revising it from time to time keeping pace with the structural changes taking place in the economy. Data published by various Government agencies can be accessed through the portal www.data.gov.in. Other agencies also bring together and publish comprehensive data on some or all sectors of the economy, these publications being, the annual **Economic Survey** of the Ministry of Finance, the Budget documents, the Five-Year Plan documents, **RBI's Statistical Handbook on the Indian Economy**, its monthly bulletin, the website on the database of the Indian Economy, and the **Human Development Reports** prepared by the Planning Commission and the State Governments. Most of these data are available in electronic format – and/or on

the websites of the agencies concerned. One can usually download the published reports of the Government agencies from their respective websites free-of-charge. Various agencies also give anonymized data for analysis purposes. The **CMIE (EIU), Mumbai** and the **EPWRF, Mumbai** also bring together and publish comprehensive data on different sectors/aspects of the economy regularly. The **Statistical Handbooks of the State/Union Territory DES** and the **District Census Handbooks** published by the Census Directorates/Registrar general of India provide comprehensive information across sectors at the State and sub-State levels. Finally, the website of the United Nations Statistics Division provide access to methodological documents, data and report of various UN organisations which can be used for cross-country comparisons. The search for any kind of data can thus very well start from one of these sources and move on, if necessary, to the primary source.

National Accounts Statistics (NAS), which consists of estimates of National Income (NNI), GNI, GDP, NDP and related macroeconomic aggregates like Consumption Expenditure, Personal Disposable Income and Factor Incomes constitute an important tool for analyzing the performance of an economy. These compiled by **CSO, MoSPI** and published every year. In compiling these, CSO follows the guidelines given in 1993 **SNA of the United Nations** to the extent feasible. These estimates are published in great detail in CSO's annual publication **National Accounts Statistics (NAS)**. The latest issue of this publication – **NAS 2014** – is also available on the **MoSPI website**. In addition, Advance Estimates of GNI, GDP, NNI and per capita income for a financial year are released two months before the end of the financial year through a **Press Note** and in the **MoSPI website**. First Revised Estimates of these aggregates relating to a financial year and revised estimates of earlier years are released by January of the next year, that is, within ten months after the close of the financial year through a **Press Note** and on the **MoSPI website**. Quarterly Estimates of GDP are also prepared and released, generally within 45 days from the close of a quarter.

NAS estimates are presented in current and constant (base year) prices. **CSO** changes the base year from time to time to take note of structural changes in the economy and to depict a true picture of the economy. The base year was changed to 2004-05 in January, 2010 and to 2011-12 in January, 2015. Thus, as of the 1st March, 2015, Advance Estimates of GDP, etc., for 2014-15, First Revised Estimates of GDP, etc., for 2013-14, estimates of GDP for the quarter ending December, 2014 and for the period April-December, 2014, all in current and constant (2011-12) prices, have been released.

Estimates of Saving and Capital Formation form part of the National Accounts Statistics. These estimates are presented in detail in the publication **National Accounts Statistics, published in May of each year**. The press note of January, 2015, has provided the estimates referred to above with a changed base year 2011-12. The detailed estimates presented in **NAS** enable an analysis of various aspects of Saving and Capital Formation like the structure of Saving and Capital Formation, the manner in which Capital Formation is financed, the role of the public sector in Capital Formation and trends in the rate of Saving and Capital Formation and net capital flow from abroad.

Estimates of State Income are made and released by **DES of all States and Union Territories except Dadra & Nagar Haveli, Daman & Diu and Lakshadweep**. Estimates of GFCF are being made by 14 state DESs. District

Income estimates are made and released by DES of States. Estimates of State Income are also available in **MoSPI website**, besides the **Economics Survey**, **RBI Statistical Handbook on the Indian Economy** and the **CMIE** and **EPWRF (State Income)** publications referred to earlier. The last publication also give the State level GFCF estimates. The **RBI Handbook**, the **Economic Survey**, the **EPWRF publication on NAS** and the **CMIE** publication referred to earlier also present the national level estimates of GDP, etc., estimates of Saving and Capital Formation made by CSO. The **CMIE** and the **EPWRF** publications also contain almost as much detail as the publication **NAS** and in a time series format.

We have also noted the limitations and inadequacies to which estimates of national and state income and those of capital formation and saving are subject. Improvement in methodology and efforts to fill gaps in data constitute a continuous process.

21.7 EXERCISES

- 1) What type of data will you need to assess the performance of Indian Economy? Explain the various sources of such data?
- 2) State the difference between “Gross” and “Net” in the context of domestic and national income. How can one arrive at “Net National Income” from “Gross Value Added at basic prices”?
- 3) What are the relationships depicted in Input-Output Table? How are these tables useful in research work?
- 4) With the help of NAS 2014, explain the structure of saving in India.
- 5) What do you mean by the term capital formation? What is the capital stock available in India?

21.8 SOME USEFUL BOOKS

Central Statistics Office (2012): *National Income Statistics – Sources and Methods*, Ministry of Statistics and Programme Implementation, Government of India, New Delhi.

_____ **(2008):** *Methods of estimating State and District Domestic Product*.

_____ **(2014):** *National Accounts Statistics, 2014*,

_____ **(2007):** *Input Output Transactions Table 2007-08*

_____ **(2015):** *Press Note New Series estimates of national income, consumption expenditure, saving and capital formation (base year 2011-12)*

International Labour Organisation (ILO) (1993) : *Resolution Concerning Statistics of Employment in the Informal Sector*, 15th International Conference of Labour Statisticians (ICLS), January, 1993. Geneva.

Indian Association for Research in National Income and Wealth & Institute of Economic Growth (1998): Golden Jubilee Seminar on Data Base of the Indian Economy, Delhi. A look at the Bulletins of the Association would also be useful.

<http://mospi.gov.in> : website of the Ministry of Statistics and Programme Implementation, Govt. of India.

<http://rbi.org.in> : website of the Reserve Bank of India.

<http://unstats.un.org> : website of the United Nation's Statistics Division.

Katyal, R.P., Sardana, M.G. and Satyanarayana, J. (2001): Estimates of DDP, Discussion Paper 2, National Workshop on State HDRs and the Estimation of District Income Poverty Under the State HDR Project Executed by the Planning Commission (GOI) with UNDP Support held in Bangalore in July, 2001, UNDP, New Delhi.

Ministry of Statistics & Programme Implementation (2001): Report of the National Statistical Commission, Ministry of Statistics and Programme Implementation, New Delhi.

System of National Accounts 1993 (SNA 1993) Commission of the European Communities, International Monetary Fund, Organisation for Economic Cooperation and Development United Nations, World Bank, Brussels/Luxembourg, New York, Paris, Washington, D.C. www.mospi.nic.in : Website of the Ministry of Statistics and Programme Implementation.

21.9 ANSWERS ON HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Coordination of statistical activities in the country maintenance of statistical norms and standards, providing liaison with control, state and international statistical agencies.
- 2) See Section 21.2
- 3) See Section 21.2
- 4) See Section 21.2

Check Your Progress 2

- 1) The data like estimates of national income and related macro economic aggregates (for example Personal Disposable Incomes, Saving, Consumption, Capital Formation etc.) which provide a comprehensive view of the internal and external transactions of the economy over a period form the system of National Accounts.
- 2) 1948-49, 1960-61, 1970-71, 1980-81, 1993-94, 1999-2000.
- 3) RBI – The Handbook of Statistics on the Indian Economy, CMIE – National Income Statistics, EPWRF publication on NAS and SDP etc.
- 4) Directorates of Statistics and Economics.

Check Your Progress 3

- 1) See Section 21.4
- 2) The annual document of NAS published by CSO.
- 3) See Section 21.5

UNIT 22 AGRICULTURAL AND INDUSTRIAL DATA

Structure

22.0 Objectives

22.1 Introduction

22.2 Agricultural Data

22.2.1 Agricultural Census

22.2.2 Studies on Cost of Cultivation

22.2.3 Annual Estimates of Crop Production

22.2.4 Livestock Census

22.2.5 Data on Production of Major Livestock Products

22.2.6 Agricultural Statistics at a Glance (ASG)

22.2.7 Another Source of Data on Irrigation

22.2.8 Other Data on the Agricultural Sector

22.3 Industrial Data

22.3.1 Data Sources Covering the Entire Industrial Sector

22.3.2 Factory Sector – Annual Survey of Industries (ASI)

22.3.3 Monthly Production of Selected Industries and Index of Industrial Production (IIP)

22.3.4 Data on Unorganised Industrial Sector

22.3.5 Industrial Credit and Finance

22.3.6 Contribution of Industrial Sector to GDP

22.4 Let Us Sum Up

22.5 Exercises

22.6 Some Useful Books

22.7 Answer or Hints to Check Your Progress Exercises

22.0 OBJECTIVES

After going through this Unit, you will be able to:

- know the kind of agricultural data available;
- explain the different sources of agricultural data and different data generating agencies;
- describe the characteristics of multivariable industrial data; and
- know how to access the different types of agricultural and industrial data.

22.1 INTROCUCTION

In the previous unit, we have looked at data on macro variables like National Income, Saving and Investment, which tell us about the dimension and broad structure of the economy and the direction in which the economy is moving.